

# API Documentation

REST API reference for SpaceCDF backend

SpaceCDF -- AI-Assisted Concurrent Design Facility

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**DRAFT**

# SpaceCDF API Documentation

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Base URL: `http://localhost:8000/api`

Total Endpoints: 105+

Format: JSON (request and response)

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## Core Design

### POST /design/quick-design

Run the full design loop (20 agents, convergence in seconds).

Request:

```
{
  "requirements": {
    "name": "My Mission",
    "mission_type": "earth_observation",
    "spacecraft_class": "nano",
    "orbit": { "orbit_type": "sso", "altitude_km": 500, "inclination_deg": 97.4 },
    "payloads": [{ "name": "Imager", "mass_kg": 1.5, "power_w": 8, "data_rate_mbps": 100 }],
    "design_lifetime_years": 3
  }
}
```

Response: { converged, iterations, total\_time\_s, parameters: {id: {value, unit, domain}}, budgets, warnings, conflicts }

### POST /studies/

Create a study (persists requirements + mission need for later reference).

### GET /studies/{id}

Retrieve study state.

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## Lifecycle Services

## **POST /lifecycle/orbit-trade**

Compute orbit trade study with scored candidates.

Request: { target\_gsd\_m, target\_revisit\_days, target\_latitude\_band, aperture\_m, max\_cost\_meur, min\_lifetime\_years, mission\_type }

Response: { candidates: [{ name, altitude\_km, inclination\_deg, achievable\_gsd\_m, revisit\_days, natural\_lifetime\_years, total\_score, rank }] }

## **POST /lifecycle/class-advisor**

Recommend spacecraft class from performance/programmatic targets.

## **POST /lifecycle/mission-trade**

Evaluate space vs non-space alternatives (filtered by mission\_type, includes constellation option).

## **GET /lifecycle/spectrum/bands**

Available frequency bands filtered by mission type and license type.

Query params: mission\_type, license\_type (amateur/experimental/commercial), data\_rate\_mbps

## **POST /lifecycle/spectrum/itu-api-template**

Generate ITU Advance Publication Information filing template.

## **POST /lifecycle/spectrum/iaru-template**

Generate IARU amateur coordination request template.

## **POST /lifecycle/regulatory/rssa**

Generate Canadian RSSSA filing template.

## **POST /lifecycle/regulatory/export-assessment**

Generate export control classification assessment.

## **POST /lifecycle/regulatory/copuos-registration**

Generate UN Registration Convention Article IV template.

## **POST /lifecycle/regulatory/eol-report**

Generate end-of-life analysis report.

## **GET /lifecycle/parametric-data**

Return all parametric model data (mass fractions, cost fractions, power duty cycles, SA power tables) with source citations.

## **POST /lifecycle/duty-cycles**

Estimate power duty cycles for a mission configuration.

Request: { spacecraft\_class, mission\_type, comms\_band, eclipse\_fraction }

## **GET /lifecycle/consistency/{study\_id}**

Run full design consistency check. Returns health score, issues by category.

## **POST /lifecycle/trade-study**

Run a tabular trade study with criteria, weightings, and options.

Request: { name, criteria: [{id, name, weight, direction}], options: [{id, name, scores: {criterion\_id: value}}] }

## **GET /lifecycle/trade-templates**

List pre-built trade study templates (orbit, component, ground, architecture).

## **POST /lifecycle/constellation/design**

Design Walker delta constellation for coverage targets.

## **GET /lifecycle/beyond-leo/orbits**

List beyond-LEO orbit options (MEO, GEO, HEO, Lunar, interplanetary) with environment data.

## **POST /lifecycle/beyond-leo/transfer**

Compute transfer orbit  $\Delta V$  from LEO to target orbit.

## **POST /lifecycle/beyond-leo/dsn-link**

Compute deep-space link budget for DSN communication.

## **GET /lifecycle/requirements/generate/{study\_id}**

Generate SMART requirements from study objectives and functions.

## **POST /lifecycle/requirements/validate**

Validate a requirement against SMART criteria.

## **POST /lifecycle/requirements/check-compliance**

Check non-compliance and get resolution options.

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## **Engineering Services**

### **GET /engineering/equipment/{domain}/search**

Search KB for equipment compatible with current design.

Domains: power, aocs, link, propulsion, structure, data, thermal, integration

### **GET /engineering/equipment/needs/{study\_id}**

Determine which equipment categories are needed based on requirements.

### **POST /engineering/equipment/check-compatibility**

Check RF compatibility between transponder and antenna.

### **POST /engineering/equipment/budget-impact**

Compute live budget impact of current equipment selections.

### **GET /engineering/verification**

Run requirement verification against current design state.

### **GET /engineering/cost**

Get cost estimate for current design.

### **POST /engineering/analysis/sensitivity**

Run parametric sensitivity sweep.

### **GET /engineering/analysis/eol-curves**

Get end-of-life degradation curves.

### **POST /engineering/analysis/trade-study**

Run equipment-level trade study.

## **POST /engineering/impact-preview**

Preview downstream impact of parameter changes (dry run ? no execution).

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## **ECSS Compliance**

### **GET /ecss/phases**

List available ECSS phases.

### **GET /ecss/compliance/{phase\_id}**

Get expected DRDs and SpaceCDF coverage for a phase.

### **GET /ecss/margins/{study\_id}**

Check all budget margins against ECSS phase-appropriate policy.

### **GET /ecss/dids**

List available DID (Document Item Description) types.

### **POST /ecss/dids/{did\_type}/generate**

Generate an ECSS DID document. Types: mrd, ts, ird, semp, rmp, conops, test\_plan.

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## **Optimizer**

### **GET /optimize/config**

Get available objectives and design variables (filtered by mission\_type if provided).

Query params: mission\_type, has\_propulsion, pointing\_accuracy\_deg

### **POST /optimize/sessions/{session\_id}**

Start an optimization run (single-objective or Pareto NSGA-II).

### **GET /optimize/runs/{run\_id}**

Get optimization run status and results.

### **POST /optimize/sensitivity/{session\_id}**

Run Morris screening sensitivity analysis.

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## **Sessions (Concurrent Design)**

### **POST /sessions/**

Create a new design session.

### **GET /sessions/{id}**

Get session state.

### **GET /sessions/{id}/history**

Get edit history for a session.

### **WebSocket /ws/session/{session\_id}**

Real-time collaboration. Message types: parameter\_update, state\_update, convergence\_complete, participant\_joined/left.

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## **Positions**

### **GET /positions/**

List all engineering positions with roles and key questions.

### **GET /positions/{id}/guidance**

Get computed guidance for a position based on current design.

### **POST /positions/answers**

Save a position question answer.

## **GET /positions/answers**

Get all saved position answers.

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## **Knowledge Base**

### **GET /kb/components/{category}**

Get all components in a category.

### **GET /kb/categories**

List all component categories.

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## **Exports**

### **POST /exports/docs/{format}**

Generate design review documents (SRR, PDR, CDR).

### **POST /exports/mbse/{study\_id}**

Generate MBSE JSON export (ECSS-E-TM-10-25A style).

### **POST /exports/fsw/{study\_id}**

Generate flight software architecture (mode manager, FDIR, TC/TM).